

Report 19/05/26

Alunno: Pancrazio Giuzio

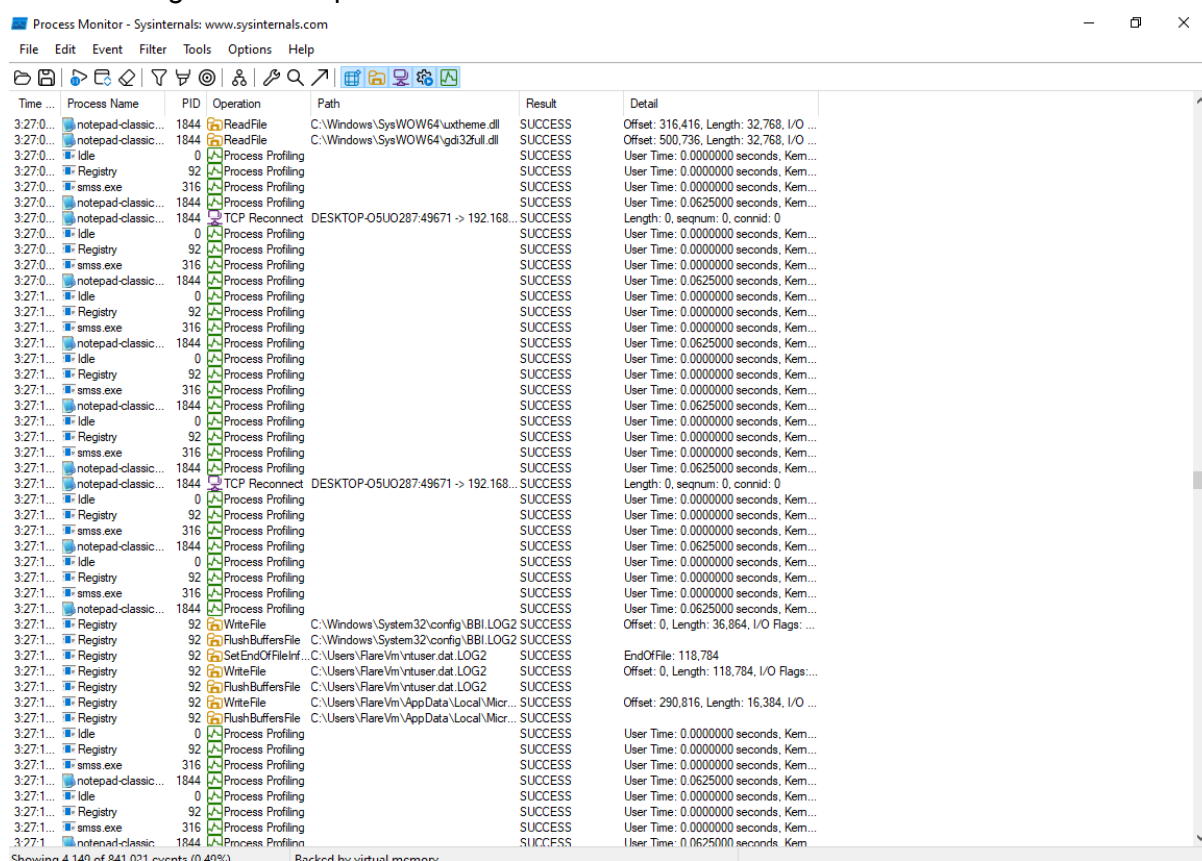
Traccia: Esercizio Traccia Rispondere ai seguenti quesiti, con riferimento al file eseguibile notepad-classico.exe contenuto nel file file zip:

<https://drive.google.com/file/d/1HNnJDSY7FbD1KHfiRzA2wVNHhZTJndUD/view?usp=sharing>

g • Identificare eventuali azioni del malware sul file system utilizzando Process Monitor, fornendo una descrizione tramite AI; • Identificare eventuali azioni del malware su processi e thread utilizzando Process Monitor, fornendo una descrizione tramite AI. Suggerimento: ChatGPT (o altri LLM) possono ricevere in input degli screenshot da

Svolgimento :

Come prima cosa andiamo a isolare la nostra flare Vm e facciamo uno snapshot della macchina , in modo tale da poterla portare allo stato precedente dopo l'esecuzione del malware notepad-classico.exe , facendo partire poi il notepad classico mentre siamo sul monitor di registro eventi possiamo notare varie cose .



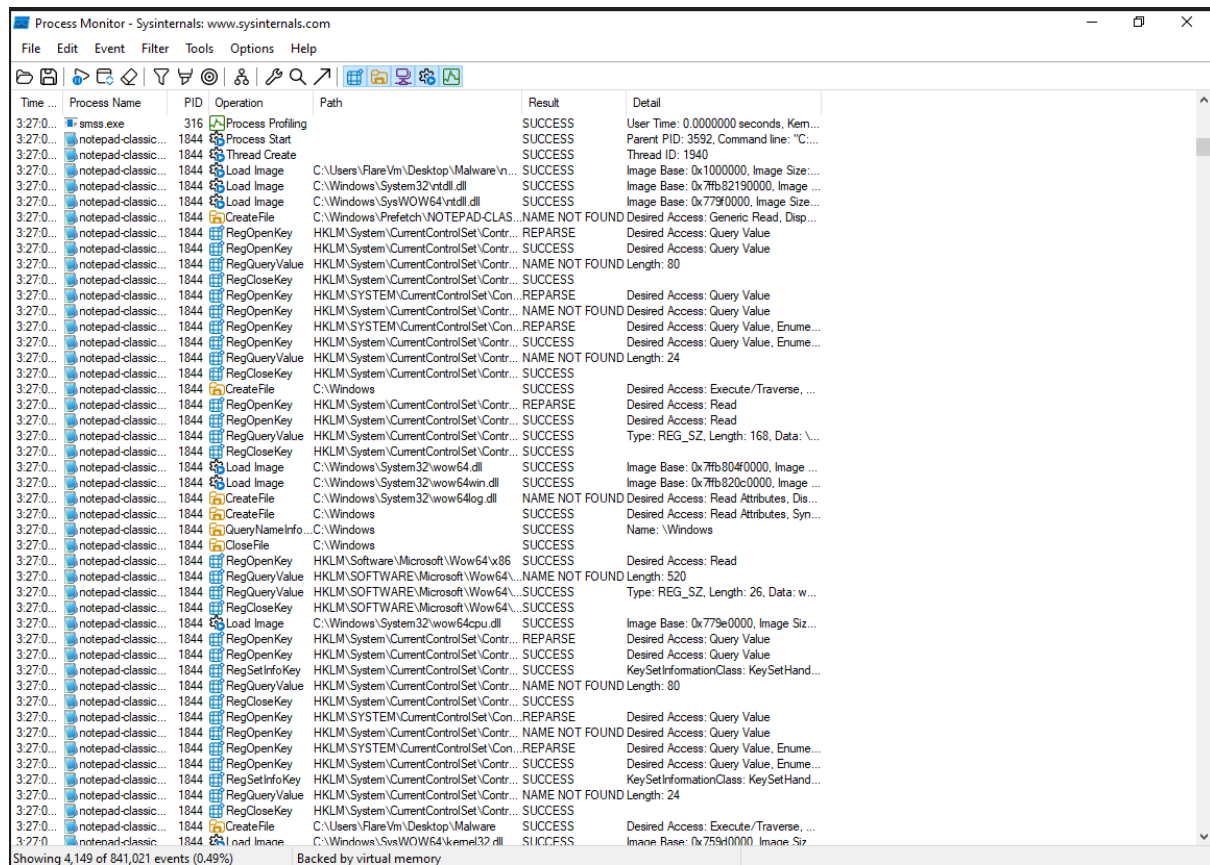
Time ...	Process Name	PID	Operation	Path	Result	Detail
3:27:0...	notepad-classic...	1844	ReadFile	C:\Windows\SysWOW64\uxtheme.dll	SUCCESS	Offset: 316,416, Length: 32,768, I/O ...
3:27:0...	notepad-classic...	1844	ReadFile	C:\Windows\SysWOW64\gd32full.dll	SUCCESS	Offset: 500,736, Length: 32,768, I/O ...
3:27:0...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:0...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:0...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:0...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...
3:27:0...	Idle	0	TCP Reconnect	DESKTOP-05UO287-49671 -> 192.168...	SUCCESS	Length: 0, seqnum: 0, connid: 0
3:27:0...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:0...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:0...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:0...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...
3:27:1...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...
3:27:1...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...
3:27:1...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...
3:27:1...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...
3:27:1...	Registry	92	WriteFile	C:\Windows\System32\config\BBI.LOG2	SUCCESS	Offset: 0, Length: 36,864, I/O Flags: ...
3:27:1...	Registry	92	FlushBuffersFile	C:\Windows\System32\config\BBI.LOG2	SUCCESS	
3:27:1...	Registry	92	SetEndOfFileInf...	C:\Users\FlareVm\ntuser.dat.LOG2	SUCCESS	EndOfFile: 118,784
3:27:1...	Registry	92	WriteFile	C:\Users\FlareVm\ntuser.dat.LOG2	SUCCESS	Offset: 0, Length: 118,784, I/O Flags...
3:27:1...	Registry	92	FlushBuffersFile	C:\Users\FlareVm\ntuser.dat.LOG2	SUCCESS	
3:27:1...	Registry	92	WriteFile	C:\Users\FlareVm\AppData\Local\Micr...	SUCCESS	Offset: 290,816, Length: 16,384, I/O ...
3:27:1...	Registry	92	FlushBuffersFile	C:\Users\FlareVm\AppData\Local\Micr...	SUCCESS	
3:27:1...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...
3:27:1...	Idle	0	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	Registry	92	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.0000000 seconds, Kem...
3:27:1...	notepad-classic...	1844	Process Profiling		SUCCESS	User Time: 0.0625000 seconds, Kem...

Un normale editor di testo non ha alcun motivo tecnico per avviare connessioni di rete (come il TCP Reconnect verso l'IP 192.168... visibile nel log). Avviando la chiamata TCP dall'interno del finto processo Blocco Note, il malware spera di ingannare i firewall locali o i sistemi di monitoraggio meno evoluti, che potrebbero avere regole più permissive per i software di sistema rispetto a un file .exe sconosciuto.

Guardando la sequenza temporale, l'attività del finto Blocco Note è strettamente alternata e sincronizzata con le continue operazioni di scrittura sul file ntuser.dat.LOG2 da parte del processo Registry. Molti malware sfruttano i file di log dell'utente o specifiche chiavi di registro per salvare configurazioni, codificare dati rubati o garantire la propria persistenza al

riavvio, "annegando" queste azioni all'interno dell'altissimo volume di scritture legittime che Windows compie ogni secondo.

In sintesi, l'azione non è solo quella di comunicare in rete, ma di farlo indossando una maschera legittima .



Time ...	Process Name	PID	Operation	Path	Result	Detail
3:27:0...	smss.exe	316	Process Profiling		SUCCESS	User Time: 0.000000 seconds, Kern...
3:27:0...	notepad-classic...	1844	Process Start		SUCCESS	Parent PID: 3592, Command line: "C:...
3:27:0...	notepad-classic...	1844	Thread Create		SUCCESS	Thread ID: 1940
3:27:0...	notepad-classic...	1844	Load Image	C:\Users\Fare\vm\Desktop\Malware\vm...	SUCCESS	Image Base: 0x1000000, Image Size...
3:27:0...	notepad-classic...	1844	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7f682190000, Image ...
3:27:0...	notepad-classic...	1844	Load Image	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Image Base: 0x7790000, Image Size...
3:27:0...	notepad-classic...	1844	CreateFile	C:\Windows\Prefetch\NOTEPAD-CLAS...	NAME NOT FOUND	Desired Access: Generic Read, Disp...
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 80
3:27:0...	notepad-classic...	1844	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Query Value, Enume...
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Query Value, Enume...
3:27:0...	notepad-classic...	1844	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
3:27:0...	notepad-classic...	1844	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
3:27:0...	notepad-classic...	1844	CreateFile	C:\Windows	SUCCESS	Desired Access: Execute/Traverse, ...
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Read
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Read
3:27:0...	notepad-classic...	1844	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Type: REG_SZ, Length: 168, Data: \...
3:27:0...	notepad-classic...	1844	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
3:27:0...	notepad-classic...	1844	Load Image	C:\Windows\System32\wow64.dll	SUCCESS	Image Base: 0x7f68040000, Image ...
3:27:0...	notepad-classic...	1844	Load Image	C:\Windows\System32\wow64win.dll	SUCCESS	Image Base: 0x7f6820c0000, Image ...
3:27:0...	notepad-classic...	1844	CreateFile	C:\Windows\System32\wow64log.dll	NAME NOT FOUND	Desired Access: Read Attributes, Dis...
3:27:0...	notepad-classic...	1844	CreateFile	C:\Windows	SUCCESS	Desired Access: Read Attributes, Syn...
3:27:0...	notepad-classic...	1844	QueryNameInfo	C:\Windows	SUCCESS	Name: \Windows
3:27:0...	notepad-classic...	1844	CloseFile	C:\Windows	SUCCESS	
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\Software\Microsoft\Wow64\w86	SUCCESS	Desired Access: Read
3:27:0...	notepad-classic...	1844	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	NAME NOT FOUND	Length: 520
3:27:0...	notepad-classic...	1844	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	Type: REG_SZ, Length: 26, Data: w...
3:27:0...	notepad-classic...	1844	RegCloseKey	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	
3:27:0...	notepad-classic...	1844	Load Image	C:\Windows\System32\wow64cpu.dll	SUCCESS	Image Base: 0x779e0000, Image Siz...
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegSetValue	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySetInformationClass: KeySetHand...
3:27:0...	notepad-classic...	1844	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 80
3:27:0...	notepad-classic...	1844	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Query Value
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Query Value, Enume...
3:27:0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Query Value, Enume...
3:27:0...	notepad-classic...	1844	RegSetInfoKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySetInformationClass: KeySetHand...
3:27:0...	notepad-classic...	1844	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
3:27:0...	notepad-classic...	1844	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
3:27:0...	notepad-classic...	1844	CreateFile	C:\Users\Fare\vm\Desktop\Malware	SUCCESS	Desired Access: Execute/Traverse, ...
3:27:0...	notepad-classic...	1844	Load Image	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Image Base: 0x759d0000, Image Siz...

Questa nuova schermata documenta la fase di ricognizione iniziale del malware, che corrisponde esattamente al momento in cui il processo viene iniettato o avviato nel sistema. All'inizio del log si nota l'operazione Process Start che sancisce la nascita di notepad-classic con il numero identificativo 1754, generato da un processo padre contrassegnato dal codice 3592.

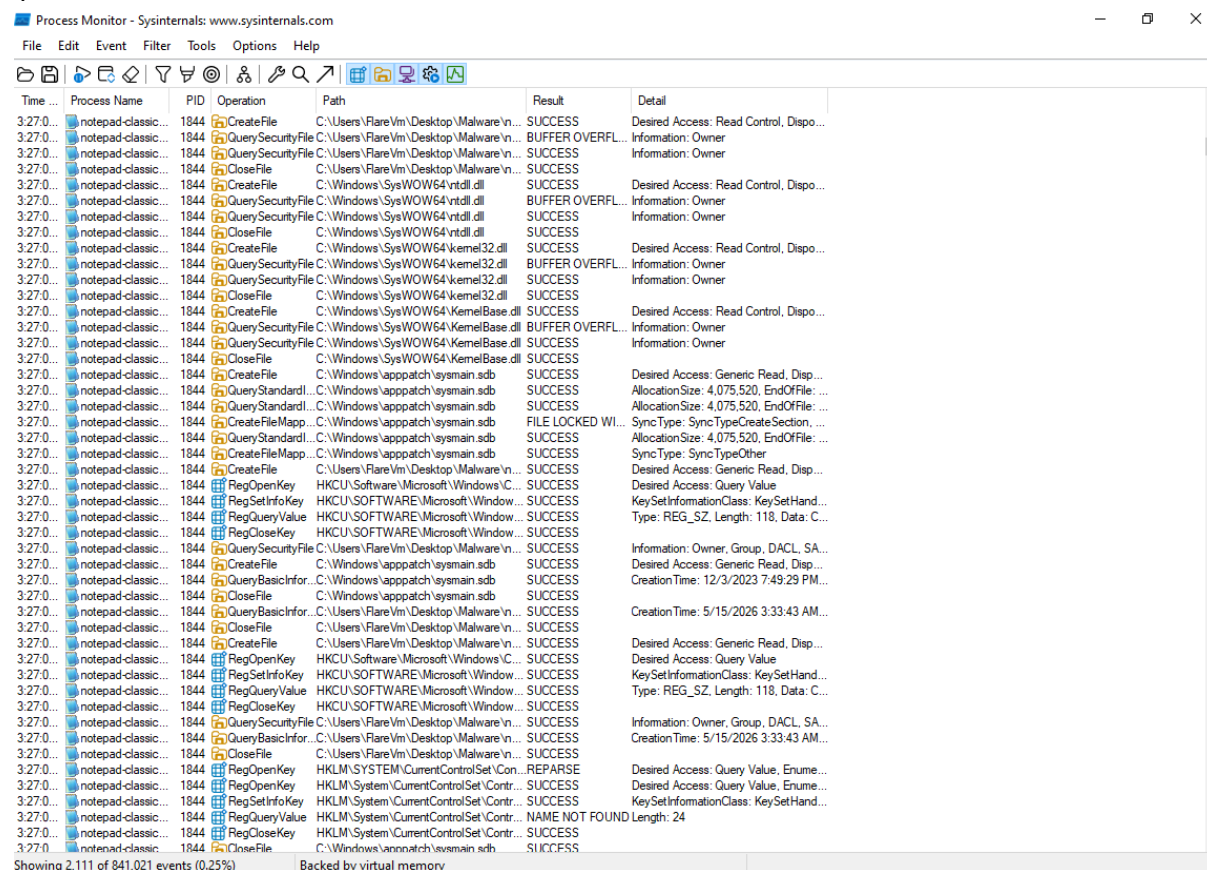
Subito dopo questa attivazione, il programma esegue l'operazione Load Image per caricare in memoria il proprio file eseguibile. Il percorso visualizzato nella colonna dei dettagli mostra che il file si trova sul desktop dell'utente all'interno di una cartella esplicitamente chiamata Malware

Nelle righe successive il processo inizia a interrogare freneticamente il Registro di sistema di Windows per mappare l'ambiente circostante. Utilizza i comandi RegOpenKey e RegQueryValue concentrandosi soprattutto sulle chiavi di controllo del sistema e sui parametri legati all'architettura a sessantaquattro bit, muovendosi tra le cartelle virtuali di Wow64 e caricando librerie cruciali come wow64cpu. Molti di questi tentativi falliscono restituendo l'errore Name Not Found, il che dimostra come il codice stia andando alla cieca alla ricerca di specifiche configurazioni di sistema, privilegi o software di sicurezza installati che potrebbero ostacolarlo.

Tra le varie ispezioni emerge anche un tentativo di accesso fallito nella cartella Prefetch di Windows, dove il sistema di solito memorizza le informazioni sull'ottimizzazione dei programmi avviati di frequente. Questo controllo serve al malware per capire se ha già lasciato tracce in passato o per prepararsi a mimetizzare la sua successiva esecuzione

automatica. Nelle ultime righe visibili, dopo aver studiato i componenti interni del sistema operativo, il programma torna a toccare la sua cartella di origine sul desktop e inizia a caricare librerie fondamentali come kernel32, che gli serviranno per sbloccare le funzioni matematiche e di memoria necessarie a far partire l'attacco vero e proprio.

Dopo aver applicato altri filtri al visualizzatore eventi Process monitor ci troviamo avanti a questo:



Time ...	Process Name	PID	Operation	Path	Result	Detail
3:27.0...	notepad-classic...	1844	CreateFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	Desired Access: Read Control, Dispo...
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Users\FlareVm\Desktop\Malware\...	BUFFER OVERFL...	Information: Owner
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	Information: Owner
3:27.0...	notepad-classic...	1844	CloseFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	
3:27.0...	notepad-classic...	1844	CreateFile	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Desired Access: Read Control, Dispo...
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Windows\SysWOW64\ntdll.dll	BUFFER OVERFL...	Information: Owner
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Information: Owner
3:27.0...	notepad-classic...	1844	CloseFile	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	
3:27.0...	notepad-classic...	1844	CreateFile	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Desired Access: Read Control, Dispo...
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Windows\SysWOW64\kernel32.dll	BUFFER OVERFL...	Information: Owner
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	Information: Owner
3:27.0...	notepad-classic...	1844	CloseFile	C:\Windows\SysWOW64\kernel32.dll	SUCCESS	
3:27.0...	notepad-classic...	1844	CreateFile	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Desired Access: Read Control, Dispo...
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Windows\SysWOW64\KernelBase.dll	BUFFER OVERFL...	Information: Owner
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	Information: Owner
3:27.0...	notepad-classic...	1844	CloseFile	C:\Windows\SysWOW64\KernelBase.dll	SUCCESS	
3:27.0...	notepad-classic...	1844	CreateFile	C:\Windows\apppatch\sysmain.sdb	SUCCESS	Desired Access: Generic Read, Disp...
3:27.0...	notepad-classic...	1844	QueryStandardI...	C:\Windows\apppatch\sysmain.sdb	SUCCESS	AllocationSize: 4,075,520, EndOfFile: ...
3:27.0...	notepad-classic...	1844	QueryStandardI...	C:\Windows\apppatch\sysmain.sdb	SUCCESS	AllocationSize: 4,075,520, EndOfFile: ...
3:27.0...	notepad-classic...	1844	CreateFileMap...	C:\Windows\apppatch\sysmain.sdb	FILE LOCKED WI...	SyncType: SyncTypeCreateSection, ...
3:27.0...	notepad-classic...	1844	QueryStandardI...	C:\Windows\apppatch\sysmain.sdb	SUCCESS	AllocationSize: 4,075,520, EndOfFile: ...
3:27.0...	notepad-classic...	1844	CreateFileMap...	C:\Windows\apppatch\sysmain.sdb	SUCCESS	SyncType: SyncTypeOther
3:27.0...	notepad-classic...	1844	CreateFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	Desired Access: Generic Read, Disp...
3:27.0...	notepad-classic...	1844	RegOpenKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Desired Access: Query Value
3:27.0...	notepad-classic...	1844	RegSetInfoKey	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	KeySetInformationClass: KeySetHand...
3:27.0...	notepad-classic...	1844	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_SZ, Length: 118, Data: C...
3:27.0...	notepad-classic...	1844	RegCloseKey	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	Information: Owner, Group, DACL, SA...
3:27.0...	notepad-classic...	1844	CreateFile	C:\Windows\apppatch\sysmain.sdb	SUCCESS	Desired Access: Generic Read, Disp...
3:27.0...	notepad-classic...	1844	QueryBasicInfor...	C:\Windows\apppatch\sysmain.sdb	SUCCESS	CreationTime: 12/3/2023 7:49:29 PM...
3:27.0...	notepad-classic...	1844	CloseFile	C:\Windows\apppatch\sysmain.sdb	SUCCESS	
3:27.0...	notepad-classic...	1844	QueryBasicInfor...	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	CreationTime: 5/15/2026 3:33:43 AM...
3:27.0...	notepad-classic...	1844	CloseFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	
3:27.0...	notepad-classic...	1844	CreateFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	Desired Access: Generic Read, Disp...
3:27.0...	notepad-classic...	1844	RegOpenKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Desired Access: Query Value
3:27.0...	notepad-classic...	1844	RegSetInfoKey	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	KeySetInformationClass: KeySetHand...
3:27.0...	notepad-classic...	1844	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_SZ, Length: 118, Data: C...
3:27.0...	notepad-classic...	1844	RegCloseKey	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	
3:27.0...	notepad-classic...	1844	QuerySecurityFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	Information: Owner, Group, DACL, SA...
3:27.0...	notepad-classic...	1844	QueryBasicInfor...	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	CreationTime: 5/15/2026 3:33:43 AM...
3:27.0...	notepad-classic...	1844	CloseFile	C:\Users\FlareVm\Desktop\Malware\...	SUCCESS	
3:27.0...	notepad-classic...	1844	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Query Value, Enume...
3:27.0...	notepad-classic...	1844	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Query Value, Enume...
3:27.0...	notepad-classic...	1844	RegSetInfoKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySetInformationClass: KeySetHand...
3:27.0...	notepad-classic...	1844	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
3:27.0...	notepad-classic...	1844	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
3:27.0...	notepad-classic...	1844	CloseFile	C:\Windows\apppatch\sysmain.sdb	SUCCESS	

Questa nuova schermata cattura un momento cruciale dell'analisi, mostrando il finto Blocco Note mentre interagisce direttamente con i database di compatibilità di Windows e, soprattutto, mentre scrive le sue chiavi di persistenza nel Registro per garantirsi l'avvio automatico.

Nelle prime righe il processo esegue operazioni di CreateFile e QuerySecurityFile sulla propria cartella nel Desktop e sulle librerie di sistema fondamentali a trentadue bit, imbattendosi ripetutamente nel risultato Buffer Overflow, che in Process Monitor indica semplicemente che il sistema ha richiesto un buffer di memoria più grande per leggere le informazioni sui proprietari dei file. Poco sotto, il malware sposta la sua attenzione sulla cartella apppatch di Windows, aprendo e leggendo ripetutamente il file sysmain.sdb, ovvero il database principale del meccanismo di compatibilità delle applicazioni, arrivando persino a generare un blocco temporaneo segnalato come File Locked With Only Readers.

La parte più allarmante dell'attività si concentra subito dopo, quando il processo inizia a interagire con la chiave di registro HKCU, che fa riferimento alle impostazioni dell'utente corrente. Il programma esegue un RegOpenKey seguito da RegSetInfoKey e RegSetValueValue puntando alla famigerata sottochiave Software Microsoft Windows CurrentVersion Run, che è il percorso standard utilizzato da Windows per avviare i programmi in automatico a ogni accesso dell'utente. I dettagli mostrano che il malware ha scritto con successo un valore di tipo stringa inserendo del codice o un percorso specifico

per fare in modo che il finto Blocco Note rimanga attivo e si riassecondi nel sistema anche dopo un riavvio della macchina.

Subito dopo aver blindato la sua presenza, il malware torna a interrogare il file sysmain.sdb per verificare le date di creazione del database di sistema, registrando i timestamp del 2023 e del 2026. Questo controllo finale serve all'eseguitibile per assicurarsi che i meccanismi di telemetria di Windows non stiano bloccando o segnalando l'eseguitibile appena registrato come una minaccia nota.

EXTRA

Traccia: Utilizzare la sandbox <https://cuckoo.cert.ee/> per analizzare l'eseguitibile notepad-classico.exe in:

<https://drive.google.com/file/d/1HNNJDSY7EbD1KHfiRzA2wVNHhzTJndUD/view?usp=sharing>

The screenshot shows the Cuckoo Sandbox web interface. The top navigation bar includes 'Dashboard', 'Recent', 'Pending', and 'Search'. The main content area is titled 'Summary' and displays analysis results for 'Archive notepad-classico.exe @ notepad-classico.zip'. A table lists various hashes and file properties. On the right, a 'Score' section indicates a score of 10 out of 10, labeled as 'very suspicious'. A 'Feedback' section is also present.

Summary	
Size	282.5KB
Type	PE32 executable (GUI) Intel 80386, for MS Windows
MD5	8a00a5c59ac157754ca575d721bcf960
SHA1	c31e260630d6553e2000f8e5f8dc270c751780d9
SHA256	d2e6c9f9273663f3218bcd7cbfb3b6f599fbce7a4ba986f9bbff77e3603988f2
SHA512	Show SHA512
CRC32	97668313
ssdeep	None

Score
This archive is **very suspicious**, with a score of **10 out of 10!**

Please notice: The scoring system is currently still in development and should be considered an *alpha* feature.

Feedback
Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Yara

- CrowdStrike_CSIT_16018_03 - Metasploit payload loader
- DebuggerCheck_QueryInfo - (no description)
- anti_dbg - Checks if being debugged
- inject_thread - Code injection with CreateRemoteThread in a remote process
- network_http - Communications over HTTP
- network_dns - Communications use DNS
- network_dga - Communication using dga
- escalate_priv - Escalade privileges
- screenshot - Take screenshot
- win_mutex - Create or check mutex

```
2026-05-19 14:20:29,046 [analyzer] DEBUG: Starting analyzer from: C:\tmp564etj
2026-05-19 14:20:29,046 [analyzer] DEBUG: Pipe server name: \\?\PIPE\OpVmecPWeFiFIDVtrIhrR
2026-05-19 14:20:29,046 [analyzer] DEBUG: Log pipe server name: \\?\PIPE\eeLDLBIDIkBgacrhMqi
2026-05-19 14:20:29,312 [analyzer] DEBUG: Started auxiliary module Curtain
2026-05-19 14:20:29,312 [analyzer] DEBUG: Started auxiliary module DbgView
2026-05-19 14:20:29,905 [analyzer] DEBUG: Started auxiliary module Disguise
2026-05-19 14:20:30,155 [analyzer] DEBUG: Loaded monitor into process with pid 508
2026-05-19 14:20:30,155 [analyzer] DEBUG: Started auxiliary module DumpTLSSMasterSecrets
2026-05-19 14:20:30,155 [analyzer] DEBUG: Started auxiliary module Human
2026-05-19 14:20:30,155 [analyzer] DEBUG: Started auxiliary module InstallCertificate
2026-05-19 14:20:30,171 [analyzer] DEBUG: Started auxiliary module Reboot
2026-05-19 14:20:30,217 [analyzer] DEBUG: Started auxiliary module RecentFiles
2026-05-19 14:20:30,217 [analyzer] DEBUG: Started auxiliary module Screenshots
2026-05-19 14:20:30,233 [analyzer] DEBUG: Started auxiliary module Sysmon
2026-05-19 14:20:30,233 [analyzer] DEBUG: Started auxiliary module LoadZeroMm0n
2026-05-19 14:20:30,375 [lib.api.process] INFO: Successfully executed process from path 'C:\Users\ADMINI-1\AppData\Local\Temp\notepad-classico.exe' with arguments '' and pid 732
2026-05-19 14:20:30,562 [analyzer] DEBUG: Loaded monitor into process with pid 732
2026-05-19 13:39:49,578 [analyzer] INFO: Analysis timeout hit, terminating analysis.
2026-05-19 13:39:49,750 [lib.api.process] ERROR: Failed to dump memory of 32-bit process with pid 732.
2026-05-19 13:39:50,125 [analyzer] INFO: Terminating remaining processes before shutdown.
2026-05-19 13:39:50,125 [lib.api.process] INFO: Successfully terminated process with pid 732.
2026-05-19 13:39:50,125 [analyzer] INFO: Analysis completed.
```

```
2026-05-19 14:35:50,707 [cuckoo.core.scheduler] DEBUG: Task #7559103: no machine available yet
2026-05-19 14:35:51,806 [cuckoo.core.scheduler] DEBUG: Task #7559103: no machine available yet
2026-05-19 14:35:52,907 [cuckoo.core.scheduler] INFO: Task #7559103: acquired machine win7x6419 (label=win7x6419)
2026-05-19 14:35:52,909 [cuckoo.core.resultserver] DEBUG: Now tracking machine 192.168.168.219 for task #7559103
2026-05-19 14:35:53,547 [cuckoo.auxiliary.sniffer] INFO: Started sniffer with PID 3516883 (interface=vboxnet0, host=192.168.168.219)
2026-05-19 14:35:53,594 [cuckoo.machinery.virtualbox] DEBUG: Starting vm win7x6419
2026-05-19 14:35:54,491 [cuckoo.machinery.virtualbox] DEBUG: Restoring virtual machine win7x6419 to vmclock
2026-05-19 14:39:11,765 [cuckoo.core.guest] INFO: Starting analysis #7559103 on guest (id=win7x6419, ip=192.168.168.219)
2026-05-19 14:39:12,771 [cuckoo.core.guest] DEBUG: win7x6419: not ready yet
2026-05-19 14:39:17,930 [cuckoo.core.guest] INFO: Guest is running Cuckoo Agent 0.10 (id=win7x6419, ip=192.168.168.219)
2026-05-19 14:39:18,040 [cuckoo.core.guest] DEBUG: Uploading analyzer to guest (id=win7x6419, ip=192.168.168.219, monitor=latest, size=6660546)
2026-05-19 14:39:19,185 [cuckoo.core.resultserver] DEBUG: Task #7559103: live log analysis.log initialized.
2026-05-19 14:39:20,279 [cuckoo.core.resultserver] DEBUG: Task #7559103 is sending a BSON stream
2026-05-19 14:39:20,694 [cuckoo.core.resultserver] DEBUG: Task #7559103 is sending a BSON stream
2026-05-19 14:39:21,508 [cuckoo.core.resultserver] DEBUG: Task #7559103: File upload for 'shots/0001.jpg'
2026-05-19 14:39:21,518 [cuckoo.core.resultserver] DEBUG: Task #7559103 uploaded file length: 89144
2026-05-19 14:39:34,353 [cuckoo.core.guest] DEBUG: win7x6419: analysis #7559103 still processing
2026-05-19 14:39:49,535 [cuckoo.core.guest] DEBUG: win7x6419: analysis #7559103 still processing
2026-05-19 14:39:49,909 [cuckoo.core.resultserver] DEBUG: Task #7559103: File upload for 'curtain/1779190789.91.curtain.log'
2026-05-19 14:39:49,912 [cuckoo.core.resultserver] DEBUG: Task #7559103 uploaded file length: 36
2026-05-19 14:39:50,107 [cuckoo.core.resultserver] DEBUG: Task #7559103: File upload for 'sysmon/1779190790.1.sysmon.xml'
2026-05-19 14:39:50,131 [cuckoo.core.resultserver] DEBUG: Task #7559103 uploaded file length: 1876178
2026-05-19 14:39:50,179 [cuckoo.core.resultserver] DEBUG: Task #7559103 had connection reset for <Context for LOG>
2026-05-19 14:39:52,548 [cuckoo.core.guest] INFO: win7x6419: analysis completed successfully
2026-05-19 14:39:52,560 [cuckoo.core.plugins] DEBUG: Stopped auxiliary module: Redsocks
2026-05-19 14:39:52,588 [cuckoo.core.plugins] DEBUG: Stopped auxiliary module: Sniffer
2026-05-19 14:39:53,939 [cuckoo.machinery.virtualbox] INFO: Successfully generated memory dump for virtual machine with label win7x6419 to path /srv/cuckoo/cwd/storage/analyses/7559103/mem
2026-05-19 14:39:53,944 [cuckoo.machinery.virtualbox] DEBUG: Stopping vm win7x6419
2026-05-19 14:42:43,082 [cuckoo.core.resultserver] DEBUG: Stopped tracking machine 192.168.168.219 for task #7559103
2026-05-19 14:42:43,612 [cuckoo.core.scheduler] DEBUG: Released database task #7559103
2026-05-19 14:42:43,663 [cuckoo.core.scheduler] INFO: Task #7559103: analysis procedure completed
```

L'intera sessione è partita dallo studio approfondito delle tracce lasciate da un eseguibile sospetto all'interno di un ambiente controllato. Attraverso il monitoraggio in tempo reale, abbiamo scoperto che il file cercava di mimetizzarsi nel sistema operativo adottando il nome ingannevole di Blocco Note, avviando al contempo chiamate di rete insolite e ispezionando freneticamente il Registro di configurazione per mappare l'architettura della macchina e tentare di blindare la propria presenza per i riavvii successivi.

I successivi report della sandbox hanno confermato in modo categorico la massima pericolosità della minaccia, identificandola come un loader per i payload del framework di attacco Metasploit. L'analisi automatizzata ha svelato la presenza di sofisticate tecniche di autodifesa, tra cui controlli per rilevare la presenza di debugger e bloccare i dump di memoria, affiancate da moduli aggressivi progettati per iniettare codice dannoso nei processi legittimi di Windows, elevare i privilegi amministrativi, scattare screenshot della vittima e comunicare con l'esterno tramite domini generati dinamicamente.

Il cerchio si chiude unendo questi comportamenti all'infrastruttura tecnologica che ha ospitato i test. I log hanno dimostrato come la sandbox abbia gestito l'intera infezione sfruttando proprio il motore di VirtualBox. Il sistema ha utilizzato i comandi automatici per ripristinare una macchina virtuale Windows a uno stato pulito partendo da un'istantanea precedente, ha catturato tutto il traffico di rete sulla scheda virtuale isolata e, al termine del tempo concesso, ha spento la macchina salvando la memoria volatile per preservare l'integrità del sistema ospitante. Questa procedura evidenzia come la corretta gestione degli snapshot sia lo strumento fondamentale per analizzare minacce informatiche così distruttive in totale sicurezza.